Structure-activity relationships generated for peptaibols produced by Trichoderma: via accelerated MD simulations

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INTRODUCTION

- *Trichoderma* fungal species play important role in agricultural and biotechnological processes.
- The members of the genus *Trichoderma* produce secondary metabolites, which include the group of peptaibols.
- They are characterized by the acetylated N-terminus and the amino-alcohol group at the C-terminus.
- Peptaibols contain non-proteinogenic amino acids in a high amount, such as α-aminoisobutyric acid (Aib) and hydroxyproline (Pro) (**Figure 1**).
- Due to their characteristic properties, peptaibols can aggregate and create ion channels in the membrane, thereby have antagonistic effects against several bacterial and fungal species.

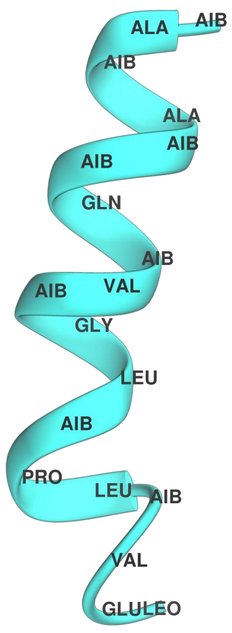

Figure 1. Peptaibol sequence

METHODS AND MATERIALS

- Peptaibol extracts were obtained from six *Trichoderma* strains from the Szeged Microbiology Collection, Szeged, Hungary (SZMC; www.szmc.hu).
- The minimum inhibitory concentration (MIC, mg ml^{-1}) values of the purified peptaibol extracts were determined against nine bacterial strains, obtained from SZMC (**Figure 2**).

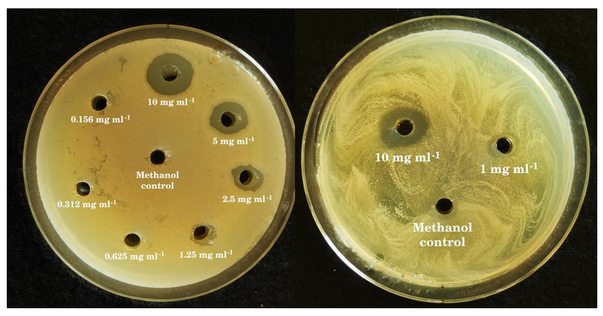

Figure 2. *T. reesei* SZMC 22616 against Gram-positive *C. michiganensis* (A), *T. longibrachiatum* f. *bissettii* SZMC 12546 against Gram-negative *R. radiobacter* (B)

Table 1. Peptaibol sequences from the *Trichoderma* strains selected for aMD simulations

Name	Sequence																			
	R	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	R18	R19	R20
Pept-A-IVa	Ac	Aib	Ala	Aib	Ala	Aib	Ala	Gln	Aib	Vxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-B-IXa	Ac	Aib	Ala	Aib	Ala	Aib	Ala	Gln	Aib	Vxx	Aib	Gly	Aib	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-A-VIa	Ac	Aib	Ala	Aib	Ala	Aib	Ala	Gln	Aib	Vxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Aib	Glu	Gln	Pheol
Pept-B-XIVa	Ac	Aib	Ala	Aib	Ala	Aib	Ala	Gln	Aib	Lxx	Aib	Gly	Aib	Aib	Pro	Vxx	Aib	Aib	Glu	Gln	Pheol
Pept-A-XVIa	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Vxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-B-XVI	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Vxx	Aib	Gly	Aib	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-A-XIXa	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Vxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Aib	Glu	Gln	Pheol
Pept-B-XIVb	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Vxx	Aib	Gly	Aib	Aib	Pro	Vxx	Aib	Aib	Glu	Gln	Pheol
Pept-A-XXIIb	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Lxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-B-XX	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Lxx	Aib	Glv	Aib	Aib	Pro	Vxx	Aib	Aib	Gln	Gln	Pheol
Pept-A-XXVb	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Lxx	Aib	Gly	Lxx	Aib	Pro	Vxx	Aib	Vxx	Gln	Gln	Pheol
Pept-B-XXVII	Ac	Aib	Ala	Aib	Ala	Aib	Aib	Gln	Aib	Lxx	Aib	Gly	Aib	Aib	Pro	Vxx	Aib	Vxx	Gln	Gln	Pheol

- The sequences contain non-standard residues Aib and Pheol, whose partial charges were calculated using the R.E.D server.
- Twelve peptaibol sequences were selected, whose unfolded conformations were prepared in water solvent (TIP3P) for aMD simulations (**Table 1**). The values of boost parameters a_1 and a_2 were chosen to be 0.16 kcal mol^{-1}, b_1 and b_2 were 4 kcal mol^{-1}.

RESULTS

Table 2. Minimum inhibitory concentration (MIC) values of the purified peptaibol extracts against Gram-positive and Gram-negative bacteria (n/a: no data)

	MIC (mg ml^{-1})	Gram-positive *Micrococcus luteus* SzMC 0197	Gram-positive *Bacillus cereus* SzMC 23292	Gram-negative *Clavibacter michiganensis* SzMC 0016	Gram-negative *Rhodococcus fascians* SzMC 21247	Gram-negative *Rhizobium radiobacter* SzMC 21407
Group A	*T. longibrachiatum* f. *bissettii* SZMC 12546	0.156	0.156	0.156	0.625	1.25
Group A	*T. longibrachiatum* SZMC 1775	2.5	1.25	0.625	5	-
Group B	*T. reesei* SZMC 22614	0.625	0.312	0.312	2.5	-
Group B	*T. reesei* SZMC 22616	1.25	1.25	1.25	1.25	-
Group B	*T. saturnisporum* SZMC 22606	2.5	0.625	0.312	0.312	-
Group B	*T. effusum* SZMC 22611	2.5	1.25	n/a	0.312	-

- The MIC values of the purified peptaibol extracts were determined against nine Gram-positive and Gram-negative bacteria using two-fold dilution series (**Table 2**).
- In the case of the purified extracts of group A strains, the strongest inhibitory effect was exerted by the extract of *T. longibrachiatum* f. *bissettii* SZMC 12546 against 4 Gram-positive bacteria, furthermore, the purified extract was also able to inhibit Gram-negative *Rhizobium radiobacter.*
- The purified peptaibols extracts from *T. reesei* SZMC 22614 and *T. reesei* SZMC 22606 strains from group B showed a strong inhibitory effect against Gram-positive bacteria.
- Peptaibol extracts from *T. reesei* SZMC 22616 strain had the lowest inhibitory effect on average against bacteria.

The most stable conformational states of sequences from group A and B were determined using the aMD simulation technique (**Figure 3**).

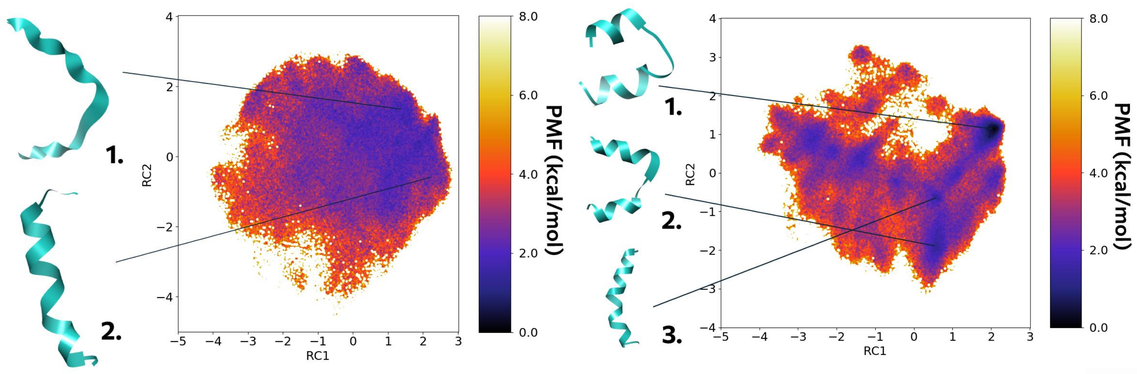

Figure 3. The three-dimensional representative models of the first three most-populated clusters in the case of Pept-B-XXVII (B) from the group B and Pept-A-IVa (A) from the group A

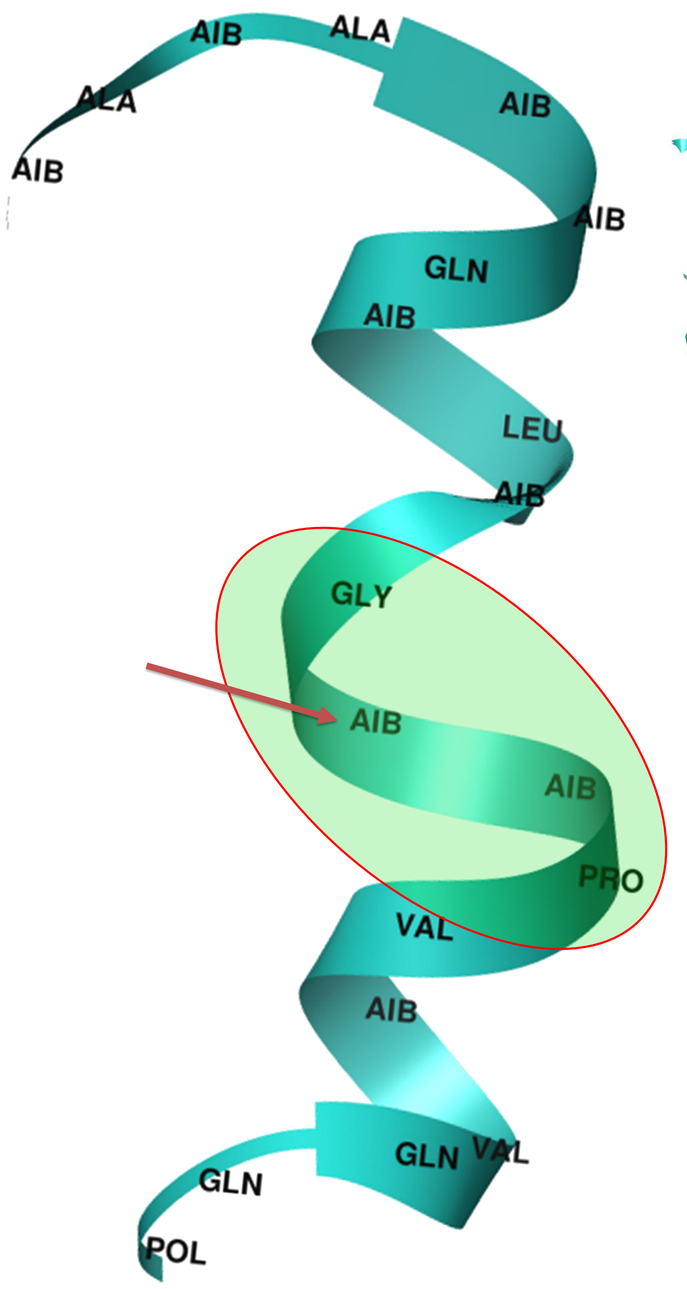

Figure 4. The most produced peptaibol (Pept-B-XXVII) from group B

CONCLUSIONS

- The difference at position R12 between the peptaibols of group A and B has an effect on protein folding: in the case of group B sequences the *Gly-Aib-Aib-Pro* motif allows a less strict folding process resulting in multiple energetically-stable intermediate states (**Figure 4**)
- In the case of *Gly-Lxx-Aib-Pro* motif of group A peptaibols where a restricted folding dynamics resulted in fewer energetically stable states (a slightly kinked linear conformation and a highly curved conformation) (**Figure 5**).
- **The restricted folding behaviour of the peptaibols from group A may promote the development of structure in membranes and the ion-channel formint ability more effectively.**

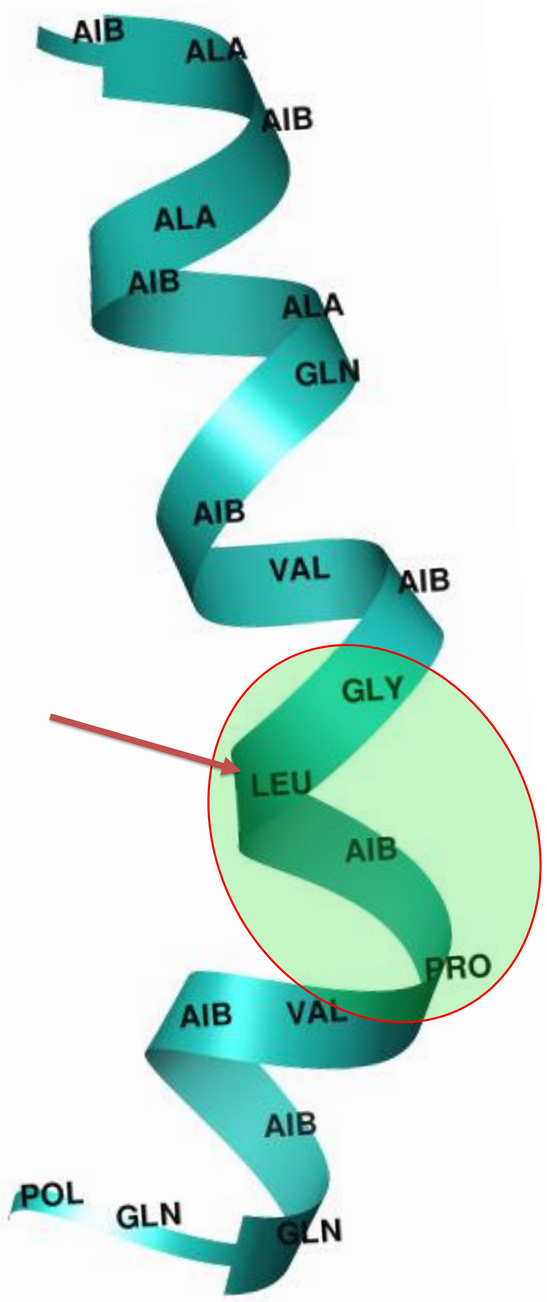

Figure 5. The most produced peptaibol (Pept-A-IVa) from group A

National Talent Programme

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